Project 2 Report — Conduct a Statistical Analysis Using Python

Program: Udacity AI Mastery Capstone (cd001), Project 2 **Dataset**: Titanic passenger manifest (n = 891) **Significance level**: alpha = 0.05, family-wise corrected (Bonferroni-Holm)

1. Project overview

The goal of Project 2 is to demonstrate fluency with classical inferential statistics in Python — pre-registering hypotheses, choosing appropriate tests, reporting effect sizes alongside p-values, and applying a multiple-testing correction across the family of tests.

The full executable analysis is in [`statistical_analysis.ipynb`](statistical_analysis.ipynb).

2. Why Titanic

The Titanic passenger manifest is the canonical dataset for inferential-statistics tutorials because a single file contains five categorical columns (`survived`, `pclass`, `sex`, `embarked`, `alone`) and five numeric columns (`age`, `sibsp`, `parch`, `fare`, `pclass`-as-ordinal) — enough variety to demonstrate every standard test on the *same* sample, which lets us discuss multiple-testing correction honestly. It is also ethically rich (women-and-children- first policy is a known *real* causal effect on the data) which makes the rubric's ethical-reflection section non-trivial.

3. Methodology

I pre-registered five two-sided hypotheses at alpha = 0.05 (table below). Each hypothesis was tested with the appropriate parametric test plus, where applicable, a non-parametric sanity check and an effect-size statistic. Levene's test was used to check the variance-homogeneity assumption of ANOVA. The family of five p-values was Bonferroni-Holm-corrected to control the family-wise error rate.

#	Hypothesis	Primary test	Effect size
1	Survival depends on sex	chi-square independence	Cramér's V
2	Survivors and non-survivors differ in mean age	Welch's t-test	Cohen's d (+ Mann-Whitney U as a sanity check)
3	Mean fare differs across `pclass`	One-way ANOVA + Tukey HSD	eta-squared (+ Levene's W)
4	`age` correlates with `fare`	Pearson r (+ Spearman ρ)	95 % Fisher-z CI
5	Survival depends on embarkation port	chi-square independence	Cramér's V

4. Cleaning decisions

- **deck** — 77 % missing; dropped without imputation (any imputation strategy would invent more data than it preserves).
- **age** — 20 % missing; median-imputed because the distribution has a mild right skew and the median is robust to that. Mean imputation was considered and rejected; an MICE imputation was not used to keep the project self-contained.
- **embarked / embark_town** — 2 rows missing; mode-imputed.

5. Results

The values below are the **exact outputs** of the executed notebook (cell 17, multipletests). Earlier drafts of this report contained a numerical error on H2 — the previous draft cited `raw p = 0.038, rejected`, which corresponds to a different random-state run and was inconsistent with the notebook. The table here matches the notebook verbatim.

#	Hypothesis	Raw p	Adjusted p	Reject H0?	Effect size
1	survived vs sex	1.20e-58	4.79e-58	**yes**	V = 0.541 (large)
2	age by survival	**0.0583**	**0.0583**	**no**	d = -0.132 (small)
3	fare across pclass	1.03e-84	5.16e-84	yes	eta^2 = 0.353 (large)
4	age vs fare correlation	3.87e-3	7.73e-3	yes	r = 0.097, 95% CI [0.031, 0.161]
5	survived vs embarked	2.30e-6	6.90e-6	yes	V = 0.171 (small-medium)

The headline finding is that **sex is the dominant survival signal** (Cramér's V = 0.541, the only large-effect result).

5.1 H2 (age by survival) — the non-rejection

Survivors had mean age 28.29 (sd 13.76, n = 342), non-survivors 30.03 (sd 12.50, n = 549). Welch's t = -1.897, raw p = 0.0583. After Bonferroni–Holm correction the adjusted p is still 0.0583 (H2 is the largest raw p in the family, so the correction does not change it). The Mann–Whitney U sanity check gives p = 0.270. Cohen's d ≈ -0.132 (small effect).

In plain language: **the survivors were on average ~1.7 years younger than the non-survivors, but at n = 891 this difference is not statistically distinguishable from chance at alpha = 0.05.** The lifeboat-first-for-children policy *is* visible in the under-12 subset but the global mean comparison is dominated by the adult majority and so the effect washes out.

5.2 H3 (fare across class) — Tukey HSD nuance

The ANOVA omnibus test rejects equality of fares across the three classes with eta^2 = 0.353 (large). The Tukey HSD post-hoc shows:

Comparison	Tukey HSD p	Significant at 0.05?
Class 1 vs Class 2	2.86e-13	yes
Class 1 vs Class 3	2.86e-13	yes
Class 2 vs Class 3	0.108	**no**

So Class 1 (First class, mean fare ≈ £84) is far above the other two, but Classes 2 and 3 (mean fares ≈ £21 and £14 respectively) are not statistically distinguishable in mean fare even though their absolute medians differ. This refinement only became visible *after* the post-hoc test — a useful reminder that an omnibus rejection does not pin down which pairwise contrasts drive it.

5.3 Variance-homogeneity caveat

Levene's test gives $W = 118.57$, $p < 1e\text{-}25$ — variances across `pclass` are very unequal. Strictly speaking ANOVA's variance-homogeneity assumption is violated; the F-statistic remains valid here only because of the large per-class sample sizes (Boneau 1960). A formal redo would use Welch's ANOVA or Kruskal–Wallis. The Tukey HSD interpretation above should therefore be regarded as approximate.

6. Interpretation for a non-technical audience

(Plain-language summary aimed at a reader who does not know what a p-value is.)

- **Whether you lived or died on the Titanic depended most on whether you were a woman.** Looking at the manifest, women had a 74% survival rate; men 19%. This is by far the strongest signal in the data, and it reflects the lifeboat-loading protocol of the time ("women and children first"), not any biological property of survival.
- **Your age made very little measurable difference.** Survivors were about 1.7 years younger on average than non-survivors, but at this sample size that difference is small enough that it could plausibly be down to luck. We cannot conclude with the data alone that young adults were systematically more likely to survive than older adults.
- **First-class passengers paid much more than the rest.** First- class fares were several times higher than second- or third-class fares on average. Second- and third-class fares were close to each other.
- **Older passengers paid slightly more on average than younger passengers.** The relationship is real but weak — knowing someone's age gives you only a small hint about their fare.
- **Where you boarded the ship was related to whether you survived,** though much of this effect is explained by the class makeup at each port (the embarkation port and class variables are correlated).

The key takeaway for a non-technical reader: most of the patterns in the Titanic data are dominated by *social* variables (class, sex, port of departure) rather than *individual* variables (age). This was not a "natural disaster" so much as a socially-stratified emergency, and the data reflects that.

7. Ethical considerations

- **Confounding and the "women-and-children-first" policy** — the large survival × sex effect is a *behavioural* confound, not a biological one. A model that uses sex as a predictor of survival would be reporting on the lifeboat-loading protocol, not on any property of the passengers. In a forward-looking ML model (e.g. an actuarial risk model) using `sex` as a feature would be both legally problematic and statistically inappropriate.
- **Sample selection bias** — the manifest reflects who boarded the Titanic, not the population of trans-Atlantic travellers in 1912; conclusions do not generalise.
- **Survivorship in the data itself** — half of the rows are reconstructed from secondary sources after the disaster; the age field in particular has a known reporting bias toward

rounded ages, which is *another* reason median imputation is more defensible than mean imputation.
- **Reproducibility ethics** — all randomness is seeded and the notebook re-executes deterministically; this is a baseline professional obligation when reporting p-values (Wasserstein & Lazar, 2016).

8. Limitations

- No multivariate analysis — each test treats the variables pairwise. A logistic-regression model of `survived` on `sex`, `pclass`, `age` would partial out the confounds and is the natural next step (Project 3 territory).
- Bonferroni-Holm is conservative for correlated tests; the effective family-wise alpha is below 0.05 for several of the hypotheses here, which would be a problem if any of the raw p-values were borderline — and **H2 is borderline**, at $p = 0.0583$. A less conservative Benjamini–Hochberg correction controlling FDR would put H2's adjusted p around 0.073, still not rejected.
- `pclass` was treated as a categorical 3-level factor for the ANOVA. Treating it as ordinal would give a different (and arguably more appropriate) trend test.
- ANOVA's variance-homogeneity assumption is violated (Levene $W = 118.57$, $p <$ 1e-25); a Welch's ANOVA or Kruskal–Wallis would be the strict-rigour alternative.

9. Future work

- Repeat every test on a held-out 20 % random split to demonstrate the asymptotic guarantees on a smaller sample.
- Compute the bootstrap 95 % CI for each effect size and report it next to the parametric one (effect sizes are more interpretable to a non-technical audience than p-values).
- Move the analysis into a logistic-regression frame in Project 3 so the *joint* effect of sex + class + age + fare can be evaluated against the marginal effects reported here.

10. References

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